

Hand Gesture Recognition

Machine learning API that predicts hand gestures from image uploads.

Directory Structure

```
hand-gesture-recognition/
├── backend/                # FastAPI backend application
│   ├── main.py            # FastAPI app with /health and /predict routes
│   ├── requirements.txt    # Python dependencies (FastAPI, Uvicorn)
│   ├── model/             # PyTorch model code and weights
│   │   ├── model.py       # Model architecture definition
│   │   └── gesture_model.pt # Pre-trained model weights
│   ├── services/          # Business logic
│   │   └── predict.py     # Prediction service implementation
│   └── schemas/           # Pydantic request/response models
│       └── predictedResponseSchema.py # API response format
├── frontend/              # Frontend application (placeholder)
├── docs/                  # Documentation and research materials
├── .env.example           # Environment variables template
└── venv/                  # Python virtual environment (git-ignored)
```

Quick Start

1. Install dependencies:

```
pip install -r backend/requirements.txt
```

2. Configure environment:

```
cp .env.example .env
```

3. Run backend:

```
cd backend && python -m uvicorn main:app --reload
```

4. Test API:

```
curl http://localhost:8000/health
```

Key Files

- **main.py** - Entry point for the FastAPI server
- **.env.example** - Template for configuration (MODEL_PATH, MODEL_DEVICE, CORS settings, file upload limits)
- **requirements.txt** - FastAPI and Uvicorn dependencies
- **gesture_model.pt** - Pre-trained PyTorch model for inference

Architecture

The API accepts image uploads and returns predicted gesture class with confidence score via the `/predict` endpoint.